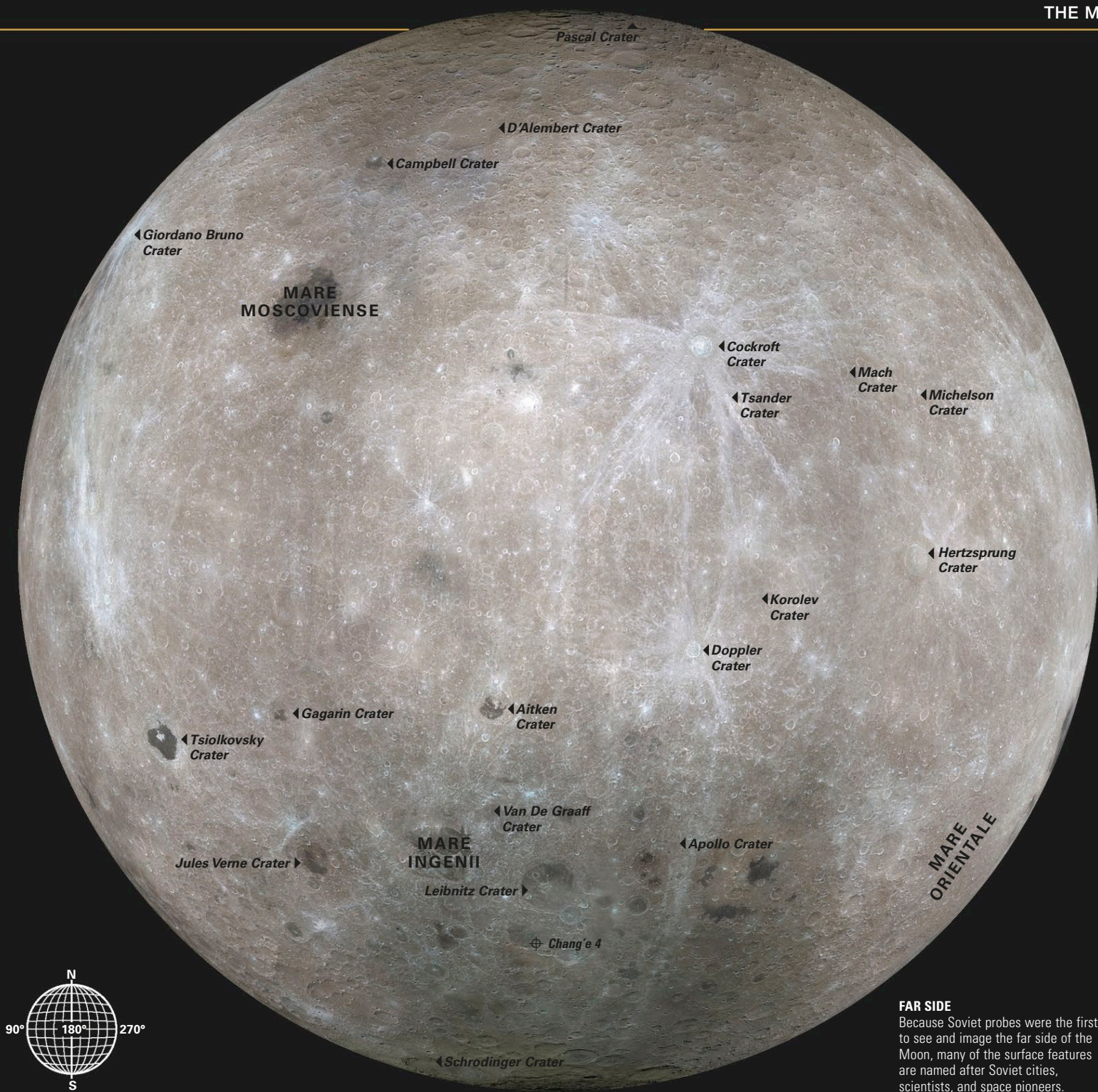




FAR SIDE
Because Soviet probes were the first to see and image the far side of the Moon, many of the surface features are named after Soviet cities, scientists, and space pioneers.

SIGNIFICANT LANDINGS ON THE MOON

Between them, automated space probes and human explorers have studied a wide range of terrains on the near side of the Moon. At first, just crashing a probe into the Moon at all was a significant achievement, but by the time of the Apollo missions, landings were targeting particular areas to answer specific questions about the Moon's geology and history.

MISSION	DATE OF ARRIVAL	TYPE	ACHIEVEMENT
Luna 2 (USSR)	September 13, 1959	Impact	Makes first crash-landing on the Moon
Ranger 7 (USA)	July 31, 1964	Impact	Takes first close-up photos of surface
Ranger 8 (USA)	February 20, 1965	Impact	Takes 7,137 good-quality photos
Luna 9 (USSR)	February 3, 1966	Lander	Makes first soft landing
Surveyor 1 (USA)	June 2, 1966	Lander	Measures radar reflectivity of surface
Luna 13 (USSR)	December 24, 1966	Lander	Successfully uses mechanical soil probe
Surveyor 3 (USA)	April 20, 1967	Lander	Images future Apollo 12 landing site
Apollo 11 (USA)	July 20, 1969	Manned	Lands first astronauts on the Moon
Apollo 12 (USA)	November 19, 1969	Manned	Makes first pinpoint landing
Luna 16 (USSR)	September 20, 1970	Lander	Makes first automated sample return

Luna 17 (USSR)	November 17, 1970	Rover	Carries first robotic lunar rover
Apollo 14 (USA)	February 5, 1971	Manned	Carries "lunar cart" for sample collection
Apollo 15 (USA)	July 30, 1971	Manned	Carries first manned lunar rovers
Luna 20 (USSR)	February 21, 1972	Lander	Makes automated sample return
Apollo 16 (USA)	April 21, 1972	Manned	Explores central highlands
Apollo 17 (USA)	December 11, 1972	Manned	Makes longest stay on Moon (75 hours)
Luna 21 (USSR)	January 15, 1973	Rover	Explores Posidonius Crater
Luna 24 (USSR)	August 14, 1976	Lander	Returns sample from Mare Crisium
Hiten (Japan)	April 10, 1993	Impact	Crashes into Furnerius region
Lunar Prospector (USA)	July 31, 1999	Impact	Orbiter makes controlled crash near the south pole to look for evidence of water
SMART-1 (ESA)	November 14, 2006	Impact	Simulates a meteor impact with crash
Chandrayaan-1 (India)	November 14, 2008	Impact	Finds evidence of water
LCROSS (USA)	October 9, 2009	Impact	Finds evidence of water
Chang'e 3 (China)	December 14, 2013	Rover	Lander deployed Yutu rover
Chang'e 4 (China)	January 3, 2019	Rover	First landing on the far side; Yutu 2 rover deployed